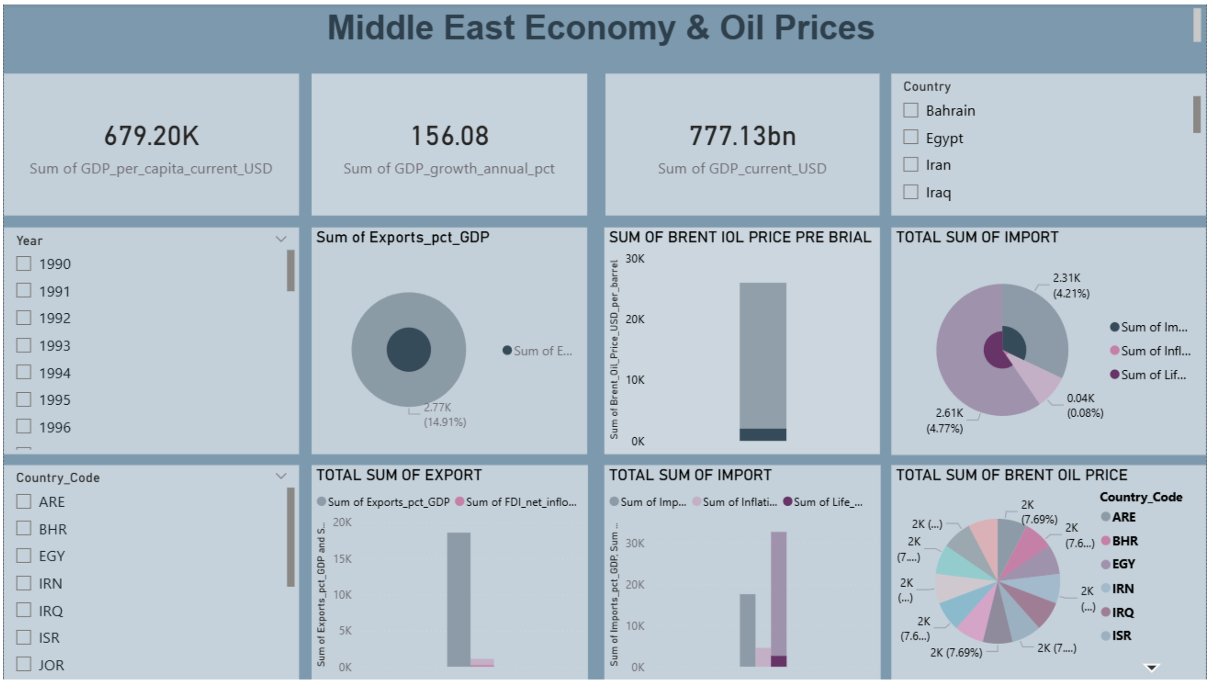
MIDDLE EAST ECONOMY & OIL PRICES DASHBOARD ANALYSIS

ABSTRACT

Economic growth in the Middle East is strongly influenced by oil production and global oil price fluctuations. This project analyzes the relationship between oil prices and economic indicators such as GDP, exports, imports, inflation, and life expectancy.

The dataset covering **1990–2024** was analyzed and visualized using **Microsoft Power BI**. The dashboard provides an interactive platform to study economic performance across several Middle Eastern countries. The visualizations help identify trends, patterns, and relationships between oil prices and economic development.



INTRODUCTION

The Middle East region is one of the world's largest producers of crude oil. Oil exports significantly contribute to national income and economic growth. Changes in global oil prices can impact government revenue, trade balance, and overall economic performance.

This project focuses on analyzing the economic indicators of Middle Eastern countries and how oil price fluctuations affect them. Using **Microsoft Power BI**, the dataset was transformed into an interactive dashboard that provides meaningful insights.

OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To analyze the economic performance of Middle Eastern countries.
- To study the relationship between Brent crude oil prices and GDP growth.
- To visualize imports and exports trends.
- To compare economic indicators among different countries.
- To develop an interactive dashboard for data-driven insights.

Tools and Technologies Used

Tool	Purpose
Microsoft Power BI	Dashboard design and visualization
Microsoft Excel	Data preparation and cleaning
Dataset	Economic indicators and oil prices data

Dataset Description

The dataset used in this project contains economic and oil price data for several Middle Eastern countries from **1990 to 2024**.

Countries Included

- Bahrain
- Egypt
- Iran
- Iraq
- Israel
- Jordan
- United Arab Emirates

Key Attributes in Dataset

- GDP per capita (current USD)
- GDP growth annual percentage
- Current GDP
- Exports (% of GDP)
- Imports
- Inflation rate
- Life expectancy
- Brent oil price (USD per barrel)

These variables help analyze the economic structure and performance of each country.

Data Preprocessing

Before creating the dashboard, the dataset was cleaned and prepared.

Steps involved:

1. Removing duplicate records
2. Handling missing values
3. Formatting numerical values
4. Creating calculated measures
5. Importing the dataset into **Power BI**

Dashboard Overview

The dashboard titled **“Middle East Economy & Oil Prices”** provides a visual analysis of economic indicators and oil price trends.

KPI Indicators

The dashboard includes three main Key Performance Indicators (KPIs):

- **GDP Per Capita:** 679.20K
- **GDP Growth Annual %:** 156.08
- **Total GDP:** 777.13 Billion

These KPIs provide a quick overview of the region's economic performance.

Dashboard Visualizations

1. Export Percentage of GDP

A **donut chart** visualizes the proportion of exports relative to GDP. This shows the dependency of countries on exports for economic growth.

2. Brent Oil Price Analysis

A **bar chart** represents Brent crude oil price fluctuations. Oil prices play a major role in determining the economic performance of oil-exporting countries.

3. Import Analysis

A **donut chart** shows the total imports across countries. This helps understand the trade balance and economic consumption levels.

4. Export Comparison

A **bar chart** compares exports and foreign direct investment indicators to analyze economic activity.

5. Import Comparison

Another bar chart compares imports with inflation and life expectancy indicators.

6. Country-wise Brent Oil Price Distribution

A **pie chart** shows how Brent oil price values are distributed across different country codes such as ARE, BHR, EGY, IRN, IRQ, ISR, and JOR.

Dashboard Filters

The dashboard includes interactive slicers:

Year Filter

Allows users to analyze economic indicators for specific years starting from **1990**.

Country Filter

Users can compare data across multiple Middle Eastern countries.

Country Code Filter

Provides additional filtering using country abbreviations.

These filters allow dynamic and flexible analysis.

Key Insights

From the analysis:

- Oil price changes strongly influence GDP growth in oil-dependent economies.
- Countries with high oil exports show stronger GDP contributions.
- Imports and exports vary significantly across different countries.
- Economic diversification helps stabilize growth in some countries.

Advantages of the Dashboard

- Provides clear visualization of complex economic data
- Enables quick comparison between countries
- Interactive filters improve user experience

- Helps identify economic trends and patterns.

Limitations

- The analysis depends on the accuracy of the dataset.
- Some economic factors such as political instability are not included.
- Real-time oil price data is not integrated.

Conclusion

This project demonstrates how data visualization tools can help analyze economic relationships effectively. By using **Microsoft Power BI**, complex economic datasets were transformed into an interactive dashboard.

The dashboard highlights the significant influence of oil prices on Middle Eastern economies and provides valuable insights for economic analysis and decision-making.

Future Enhancements

Future improvements may include:

- Integrating real-time oil price data
- Adding machine learning predictions
- Including additional economic indicators
- Expanding the dataset to include more countries